Differential Amplifiers Experiment 2

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Pre-LAB Work

1. For the circuit given in Fig. 1, calculate the differential mode (Vout/Vid) and common mode (Vout/Vcm) voltage gains. Ignore the Early Effect and assume that all the transistors are operating in active mode. Vcc=15V and Vee=-15V.

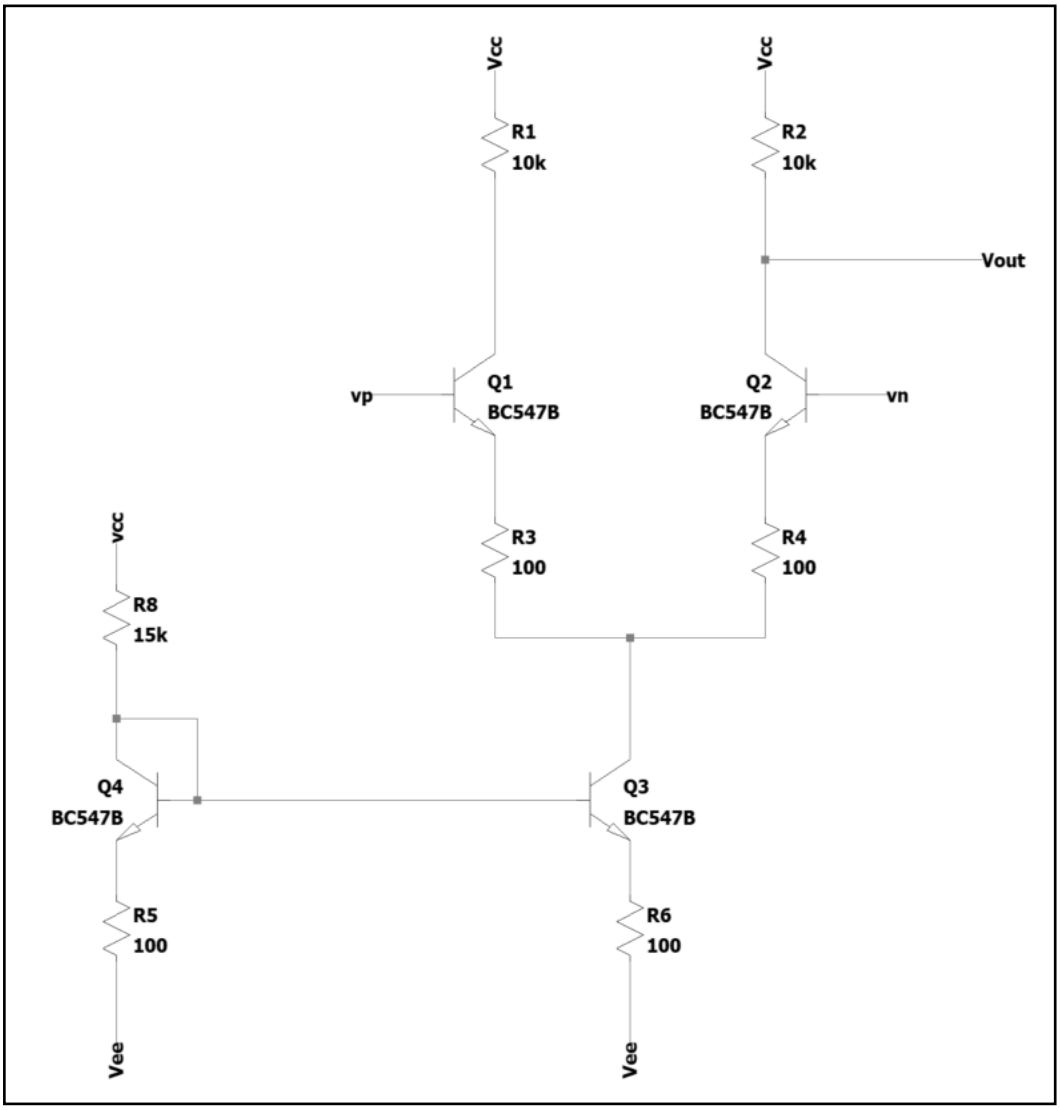


Figure 1

Answer:

2. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Apply a sine wave of 1kHz with an amplitude of 50mV to vp and connect vn to ground. Measure the differential mode voltage gains.

Ad=(Vout[peak-to-peak]/100mV)= 37.767

3. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Apply a sine wave of 1kHz with an amplitude of 50mV to both vp and vn. Measure the common mode voltage gains.

Ac=(Vout[peak-to-peak]/100mV)=0.01548

Equipment/Component List

Table 1: Equipment/Component List

Equipment/Component	**Model/Value**	**Count**
Breadboard	-	1
Resistor	100Ω	**4**
Resistor	10kΩ	**2**
Resistor	15kΩ	1
BJT (npn)	BC547	**4**
Multi-meter	-	1
Voltage Source	15V, -15V	1
Probes & Cables	-	-

Experiment Procedure

4. Adjust Vcc to 15V and Vee to -15V.
5. Set-up the circuit in Fig. 1.
6. Connect Vn to ground.
7. Apply a sine wave of 1kHz with an amplitude of 50mV to Vp.
8. Plot Vout and measure the peak-to-peak value of the output.

Plot:

9. Calculate the differential mode voltage gain.

Ad=(Vout[peak-to-peak]/100mV)=

10. Remove the ground connection from Vn and then connect Vn to Vp.
11. Calculate the common mode voltage gain.

Ac=(Vout[peak-to-peak]/100mV)=

12. Compare the calculated (1), simulated (2, 3) and measured (9, 11) common and differential mode voltage gains.

Answer:

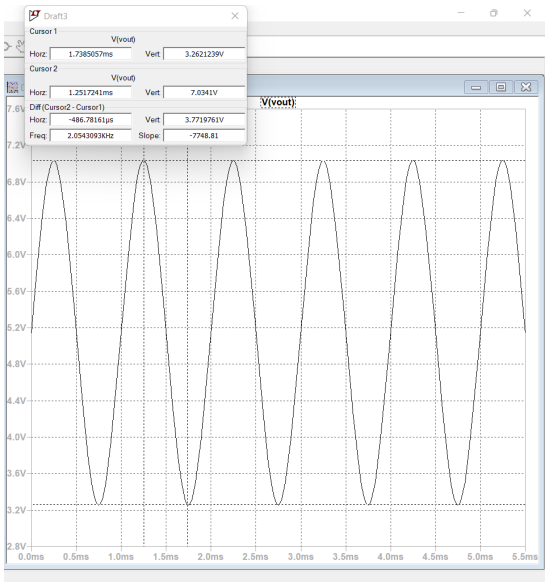


Figure 1: Ltspice Simulation result when Vn is connected to the ground

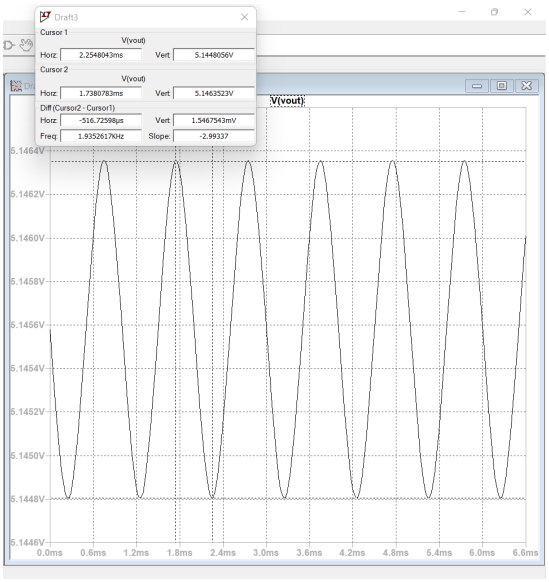


Figure 2: Ltspice Simulation result when Vn is connected to the -15V